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The Blessings of Overparameterization

Applications in Solving Economic Models

Mahdi E. Kahou[1] Jesús Fernández-Villaverde[2]
[1]Bowdoin College [2]University of Pennsylvania

The moral
The parameter count was never the risk.

Motivation

Deep learning now solves **high-dimensional** dynamic models once thought intractable. But these networks are **massively overparameterized**: far more parameters than "data" points.

The classical worry is **overfitting** and poor out-of-grid performance. One key difference from statistics: our residuals are deterministic evaluations of the functional equation, so there is **no noise to overfit**. The question is what the algorithm does **between the grid points**.

The question

When we solve dynamic models with neural networks, is overparameterization a **necessary evil**, or does it buy us something?

Setup: models as functional equations

An economic object $f : \mathcal{X} \to \mathcal{R} \subseteq \mathbb{R}^d$ (a policy, a value function) with model error $\ell(x, f)$ (Euler or Bellman residual). A **solution** satisfies $\ell(x, f^*) = \mathbf{0}$ for all x.

Pick a parametric family $f(\cdot;\theta)$, sample a grid $\mathcal{D} = \{x_1, \ldots, x_N\}$, and minimize the mean squared residual:

$$\min_{\theta} \ \mathcal{L}(\theta; \mathcal{D}) \quad = \quad \frac{1}{N} \sum_{x \in \mathcal{D}} \left\| \ell\big(x, f(\cdot;\theta)\big) \right\|_2^2$$

Neural networks are a **highly overparameterized** class ($M \gg N$). Training is **non-convex**, so the solution can depend on the **initialization**. We run **50 seeds** at every width.

What we find

- **Accuracy improves with size.** Wider networks give more accurate solutions.
- **Stability improves with size.** The spread across seeds collapses as width grows.
- **A mechanism.** In a dissectable random-features case: width buys agreement, the implicit bias decides what you agree on.

Four models of increasing difficulty

- **Linear-Quadratic:** closed form, sharpest test of error.
- **McCall job search:** 1D, kink at the reservation wage.
- **Real Business Cycle:** 2D stochastic state, no closed form, benchmarked vs. VFI.
- **Small Open Economy:** 5D, three shocks, beyond any grid benchmark.

This poster details the first two and the random-features dissection. In all four, **accuracy** and **stability** improve with overparameterization.

The two worked models

Linear-Quadratic. A price-taking firm chooses investment u at quadratic adjustment cost, state (y, Y):

$$v(y, Y) = \max_{u} \ (\alpha_0 - \alpha_1 Y)y - \tfrac{\gamma}{2}u^2 + \beta\, v(y', Y') \quad \text{s.t.} \quad y' = y + u, \qquad Y' = h_0 + h_1 Y$$

McCall job search. An unemployed worker gets i.i.d. offers $w \sim f$ on $[0, B]$ and accepts or keeps searching:

$$v(w) \quad = \quad \max\left\{ \tfrac{w}{1-\beta},\ c + \beta \int_0^B v(w') f(w')\, dw' \right\}$$

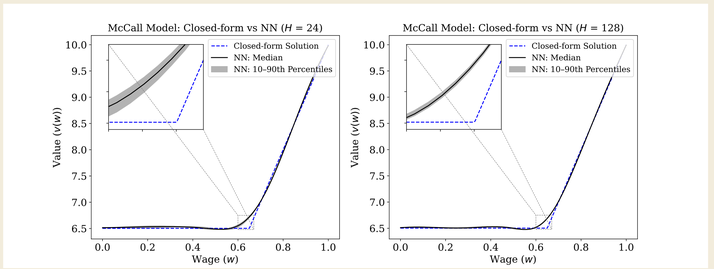
Result 1 – Linear-Quadratic regulator

A competitive firm with quadratic adjustment costs. **Closed-form** solution, so the approximation error is known exactly at every point. Minimize the Euler residual over a grid of $n_Y = 6$ points. Train and test loss stay **indistinguishable** at every width.

Median and max relative error fall to $\sim 10^{-4}$ and the across-seed band narrows: accuracy **and** agreement improve together.

Result 2 – McCall job search

Nonlinear and **non-smooth**: the value function has a **kink** at the reservation wage. The network must learn it from only $N = 11$ points. Median relative error falls from $\sim 17\%$ to $\sim 0.2\%$, its plateau set by the kink, not the network.



Narrow ($H = 24$, left): the median tracks the truth but seeds disagree near the kink. Wide ($H = 128$, right): 50 initializations **converge to the same function**, corner gently rounded. **Stability blessing confirmed.**

A careful examination: random features

Something selects one solution among many. To see what, strip the model to a case where nothing is hidden: draw the first layer $(\tilde{w}_j, \tilde{b}_j)$ once, **freeze it**, train only the readout $\theta = (v_1, \ldots, v_H, c)$:

$$f_H(Y;\theta) \quad = \quad \sum_{j=1}^{H} v_j\, \sigma(\tilde{w}_j Y \ + \ \tilde{b}_j) \ + \ c$$

The LQ residual is then **linear in** θ:

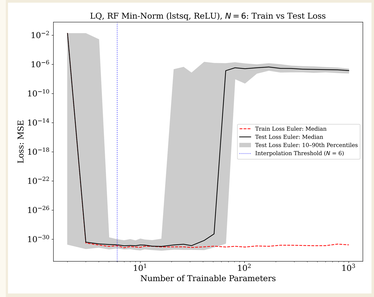
$$\Phi\, \theta \quad = \quad d, \qquad \Phi \quad \in \quad \mathbb{R}^{N \times (H+1)}$$

For $H + 1 > N$ there are **infinitely many exact solutions**. The loss cannot identify θ: **the algorithm is the identifying assumption**. Gradient descent from zero converges to the minimum-norm interpolant

$$\theta^\star \quad = \quad \operatorname*{argmin}_{\theta:\, \Phi\theta = d} \|\theta\|_2 \quad = \quad \Phi^\top (\Phi\Phi^\top)^{-1} d$$

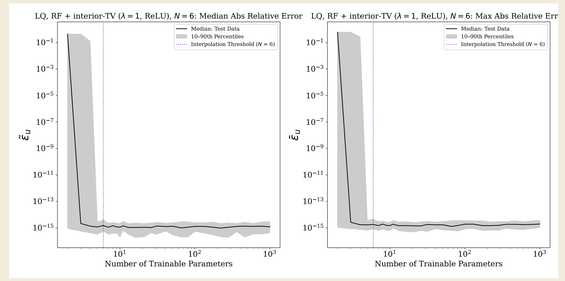
which we solve directly, **bypassing training entirely**.

Minimum norm knows no economic model



With **ReLU** features the train loss is pinned at machine zero, but the test error **rises** with width. Min-norm in **coefficients** is not smoothness in **functions**: kinked features inject curvature between grid points, where the truth is linear. That is where the error lies.

The cure: penalize the kinks between points



Charging for slope jumps inside the training range (a measure of interior curvature) snaps the fit back to the line: flat at 10^{-15} at every width.

Conclusion

Across four models (LQ, McCall, RBC, and a five-dimensional small open economy) and smooth activations, **out-of-sample residuals fall and seeds concentrate as width grows**. The mechanism: **width buys agreement; the implicit bias decides what you agree on**. The parameter count was never the risk.